

Post-TIPS OHE: Are we really making a mountain out of a molehill?

To the Editor:

We read with great interest the paper published in *Journal of Hepatology* by Nardelli and colleagues.¹ Whether caused by a bypass or not, overt hepatic encephalopathy (OHE) has long been a concern to both hepatologists and interventional radiologists. This sensitivity does not come out of thin air but derives from evidence-based results on the natural history of cirrhosis. With a multicenter observational cohort, Nardelli *et al.* presented encouraging findings that episodic OHE after TIPS does not affect survival in patients with cirrhosis. Yet before we could feel relieved about these results, some questions remained to be discussed. The core issue here is that not all OHE is the same; factors such as timing and severity of OHE might determine the way we interpret these results.

To begin with, timing should be taken into consideration. As we have previously reported,² the incidence of post-TIPS OHE peaked within 3 months after the procedure, thus, OHE within this time frame was defined as early OHE,³ the definition of which is relevant because we identified it as an independent predictor of survival with a hazard ratio of 2.75. Additionally, when early OHE was further stratified into early-recurrent and single OHE, the reduced survival was mainly due to the patients with early recurrent OHE ($p < 0.001$) while no significant difference in survival was observed in the patients with a single early episode ($p = 0.24$). This conclusion was recognized by the 2022 EASL Clinical Practice Guidelines on the management of HE.⁴ The current study by Nardelli *et al.* is derived from a cohort with an undivided follow-up period of 30 months. From this perspective, we are curious as to whether OHE within or beyond 3 months after TIPS influences survival differently or whether post-TIPS OHE simply does not influence survival at all regardless of timing in this cohort.

Secondly, various severities of OHE might influence survival differently. It is intuitive to speculate that different degrees of OHE, with different degrees of damage to the central nervous system and multiple organs, have differing prognoses. Indeed, a retrospective study⁵ that included 494 patients with end-stage liver disease, of whom 223 patients underwent TIPS, showed that HE of grade III or higher was associated with an increased risk of death (hazard ratio = 3.7, 95% CI 1.9–7.3, $p < 0.01$); there was a demonstrable correlation between severity

of HE and survival. Another observational study also confirmed that patients with grade III–IV HE had significantly poorer survival than those without,⁶ which was consistent with our clinical observation. Therefore, providing more detailed information on the relationship between the severity of OHE and survival should be helpful to refine the major conclusions of this study.

Post-TIPS OHE has long been an annoying issue, and every physician wishes to provide patients with explanations that post-TIPS OHE does not influence survival. Yet before doing so, further investigations are needed. The more we are concerned about OHE, the more careful we must be when facing it. Just like Nardelli *et al.* mentioned, even if survival remains unaffected by OHE, intensive research is still warranted since our pursuit should not simply focus on survival, but also on enhancing the quality of life of patients.

Xing Wang^{1,†}

Qiuhe Wang^{2,†}

Zhengyu Wang¹

Bohan Luo¹

Guohong Han^{1,*}

¹Department of Liver Diseases and Interventional Radiology, Digestive Diseases Hospital, Xi'an International Medical Center Hospital, Northwest University, Xi'an, China

²Department of Cardiology, Tangdu Hospital, Airforce Medical University, Xi'an, China

*Corresponding author. Address: Department of Liver Diseases and Digestive Interventional Radiology, Xi'an International Medical Center Hospital of Digestive Diseases, Northwest University, No. 777 Xitai Road, High-tech Zone, Xi'an 710100, China. Tel.: +86-29-68302656, fax: +86-29-68302656.

E-mail address: hangh@fmmu.edu.cn (G. Han)

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[†]These authors contributed equally to this work.



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Conflict of interest

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Authors' contributions

Drafting of the manuscript: WX,WQH,WZY,LBH. Critical revision of the manuscript for important intellectual content: HGH.

Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.



Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.03.026>.

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